pynxtools plugin audit

Inventory of the readers, parsers and format handlers in the pynxtools ecosystem

TASK DESCRIPTION	
Goal	Systematically analyse the pynxtools ecosystem to identify and document the individual readers, parsers and format handlers implemented in each plugin.
What is examined (for each handler)	• Supported input formats • Data-processing procedures • Unit conversions • Mathematical transformations • Metadata-mapping rules • Assumptions about data structure and interpretation
Comparison	The implemented behaviour is compared against the relevant NeXus application definitions, data-format specifications and documented conventions, to identify non-standard processing, implementation-specific deviations and implicit assumptions.
How findings are recorded	Each finding goes into a structured table with: • plugin and handler names • source-code repository • relevant documentation and implementation locations • a description of the identified behaviour • supporting evidence from source code, tests or technical documentation
Result	A traceable overview of format-specific processing that makes it easier to assess consistency, interoperability and compliance with the intended NeXus data model.

GENERAL DOCUMENTATION	
Tutorial	Build a reader
How-to	Build a plugin
How-to	Use the pynxtools test framework
How-to	Use the multi-format reader
How-to	Use pynxtools with NOMAD
Background	pynxtools architecture

HOW TO READ THIS WORKBOOK	
pynxtools_plugins_general	Overview: one row per plugin with state, technique, NeXus application definition(s) and links. Plugin names link to their detail sheet.
pynxtools-xps	Filled in: parser-by-parser analysis (standard behaviour, special behaviour, formulas, notes on possible issues).
pynxtools-xrd	Template: general table and detailed table, no entries yet.
Other plugin sheets	Template: empty tables with headers, ready to be filled (apm, ellips, em, igor, mpes, microstructure, raman, spm, stm, xas, multiformat reader).

LEGEND	
Public	Released and publicly available
Public (under development)	Publicly available, still being developed
Public (archived)	Publicly available but archived / no longer maintained
Under development	Not yet released
Private	Private repository
—	None / not applicable
Blue underlined text	Clickable link (GitHub, documentation, source file, NeXus definition, or another sheet)